

Warehouse Layout Optimization: A Simulation Study of Storage Assignment Policies at the Schneider Electric Manisa Plant Warehouse

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Abstract—This study presents a discrete-event simulation model built to evaluate storage-assignment policies at the Schneider Electric Manisa medium-voltage switchgear plant, with the goal of improving the efficiency of its kitting operation. The work was motivated by a limitation in the plant’s existing ABC/FMR classification tool, which ranks materials by aggregate demand and ignores which production line a material is kitted for. As a result, items that belong to the same kit are scattered across different corridors, which creates queues at the shared reach-truck (RT) fleet and delays material supply. A SimPy-based model was developed from four months of historical SAP dispatch data and calibrated with a video time-motion study, and six storage-assignment policies were compared across five key performance indicators: kitting lead time, operator waiting time, reach-truck utilization, throughput, and operator walking distance. Across $N = 20$ paired replications, the proposed *Line-aware Slotting* policy reduced mean kitting lead time from 12.37 to 6.91 minutes per order, a reduction of about 44% ($p < 0.001$), and lowered reach-truck utilization from 21.7% to 1.3%. Walking distance rose only slightly, from 89.0 to 92.4 metres per order, which shows that the improvement comes from removing reach-truck calls from the kit path rather than from making operators walk less. One-way ANOVA and common-random-number paired t -tests confirmed that these differences are statistically significant and not an artefact of simulation noise. The study shows that a meaningful gain is available from re-slotting alone, without new equipment or physical layout changes.

Index Terms—discrete-event simulation, storage location assignment, warehouse slotting, kitting, reach-truck utilization, common random numbers

I. INTRODUCTION

Warehouse optimization is the systematic improvement of facility operations toward efficient, flexible, and cost-effective performance. It draws on several levers, including workflow design, technology, spatial layout, and inventory management, and its aim is to reduce operating cost while improving speed, accuracy, and productivity. Among warehouse activities, order picking is the most labour-intensive and costly process, and roughly half of all picking time is spent travelling between storage locations [9]. Storage location assignment, that is deciding which material is placed in which bin, is therefore one of the few levers that can improve picking efficiency without significant capital expenditure [5].

Schneider Electric is a global company in energy management and automation with operations in more than 100

countries. Its Manisa medium-voltage switchgear plant runs a pallet-rack warehouse that feeds several parallel production lines through a coordinated kitting process. Kitting here means assembling the complete set of components for a specific production order, a kit, before assembly begins. The warehouse therefore dispatches coordinated batches of components, one batch per order, rather than single parts.

The facility handles both pallet-based and box-based materials. It uses multi-level rack systems, dedicated fast-mover shelves, and kitting corridors that feed the production lines. Material flow is maintained by a milkrun (waterspider) system on fixed 45-minute cycles, using reach trucks (RT) and electric pallet trucks (EPT). These cycles impose tight time limits with little tolerance for delay. The warehouse also operates under strict safety rules, including unidirectional flow and dedicated kit paths, which restrict routing options and increase the impact of slotting decisions.

The scale of the operation makes storage assignment important. The warehouse processes roughly 400 kit orders on an active working day; the SAP ZWM92 dispatch log covering 2 January to 18 May 2026 records 40,804 kit orders across 103 active days. At this volume, a one-minute reduction in mean kitting lead time saves the equivalent of more than six working hours of aggregate delay each day.

Despite these stakes, the current storage-assignment practice, driven by a capacity heuristic and a traditional ABC/FMR classification, does not account for the production line a material is kitted to. High-frequency items are spread across different corridors, which produces queues at the reach-truck fleet and delays material supply. Physically re-slotting thousands of pallet positions to test alternatives would be expensive and disruptive. This study therefore develops a validated discrete-event simulation (DES) model to evaluate candidate dependency-aware slotting policies against four months of real demand, so their operational impact can be measured without moving a single physical pallet.

A. Problem Statement

The warehouse at the Manisa plant operates in a dynamic environment where pallet storage, kitting, and internal transport happen at the same time. Several operational and

capacity issues in the current structure limit its scalability and efficiency.

The facility is a multi-level rack system built mainly for standard Euro pallets (80×120 cm), with a capacity of about 3,000 pallet positions distributed across several modules, and each rack section holds three pallets. The layout is organized into parallel aisles about three metres wide, where handling is carried out mostly by reach trucks. This narrow-aisle arrangement creates a strong dependency on shared equipment and raises the risk of congestion. For safety, access to most standard racks is restricted to one side, with the rear closed to prevent contact between operators and vehicles. Frequently used materials are stored at ergonomically accessible levels on dedicated fast-mover racks with double-sided access, where replenishment is performed from the front by reach truck while operators pick from the rear through the kitting aisle, so picking and replenishment can proceed at the same time.

Materials enter through receiving and are assigned to storage based on stock level and operational need. High-frequency items are generally placed on fast-mover racks and less frequent items in upper levels. Retrieval for production is done either through coordinated kitting or direct pallet handling, and internal transport follows the fixed milkrun routes.

Despite this structured system, current storage and allocation policies do not reflect actual material usage across the different production lines. The existing classification uses an internal combined ABC/FMR tool, where the ABC component represents material cost and the FMR component represents usage frequency. Although the tool updates quarterly and flags materials that are no longer or rarely used, it has a structural flaw: it treats all materials as if they came from a single source and ignores the different quantities each line consumes. The raw output is therefore not reliable for direct operational use. As a workaround the plant analyses per-line kit consumption, but the overall system still assigns locations by aggregate frequency without considering line dependencies. This scatters the materials of one kit order across different rack levels and zones [6], [17], which raises coordination effort and delays material supply [17].

The layout also does not fully use the available space. Storage density could be raised through layout redesign, extra racking, or better use of vertical space, but such physical changes need quantitative assessment before they are attempted. The plant currently follows internal rules that have not been compared against alternatives under consistent conditions.

This study therefore targets the inefficiency that comes from the lack of a structured, dependency-aware approach to storage assignment. Rather than proposing immediate physical changes, it evaluates alternatives in a virtual environment, which allows a low-cost assessment of policy changes before any physical implementation. The alternatives range from established methods such as Double ABC and time-based classification to a strategy tailored to the plant, all measured against the current baseline.

B. Motivation

Warehouse efficiency is the main driver of production stability at this plant. Material flow is governed by a milkrun system on 45-minute cycles that completes nine tours per day to replenish the lines. Each station must hold enough stock for one 45-minute interval plus a 20% tolerance margin. Within these limits, even small delays in retrieval or kit preparation can exhaust the buffer and cause a line stoppage. Occupational health and safety is also a design constraint: the project aims to raise throughput while keeping strict separation between pedestrian operators and vehicles such as reach trucks and forklifts.

The facility is operationally mature and has a culture of improvement. A previous internal study removed one reach truck from the fleet without losing performance, which shows that gains are achievable through better system design rather than capital spending. The warehouse currently runs a fleet of 7 reach trucks, 3 forklifts, 7 milkrun trains, and 2 electric pallet trucks, together with Kardex units for small parts.

Even so, inefficiencies persist. The main analytical weakness is the classification tool, which treats all materials as one homogeneous group and ignores product complexity and line destination. Operationally this creates waiting waste during kit preparation: when the materials of one kit are scattered, with some on accessible fast-mover racks and others in high positions that need a reach truck, the operator in the kit corridor must wait for an RT driver. This waiting extends lead time and reduces synchronization with production.

The motivation of this study is therefore to provide a scalable, low-cost solution through a data-driven slotting approach. By combining a dependency-aware value-based (ABC) and frequency-based (FMR) analysis, the work aims to place high-frequency materials in ergonomic, lower-level positions so they can be picked by hand, removing the reach-truck dependency from the kit path.

C. Research Objectives and Scope

The objectives of this study are:

- 1) to develop and validate a SimPy-based discrete-event simulation model of the Schneider Electric Manisa warehouse, driven by four months of SAP dispatch data and calibrated with a video time-motion study;
- 2) to design and implement six storage-assignment policies, including the proposed *Line-aware Slotting* policy, within a single simulation framework that holds the demand stream and resource pool constant;
- 3) to compare the policies across five key performance indicators (kitting lead time, operator waiting time, reach-truck utilization, throughput, and operator walking distance); and
- 4) to recommend a placement strategy with statistical evidence, using one-way ANOVA, common-random-number paired *t*-tests, and 95% confidence intervals from $N = 20$ replications.

The scope is limited to storage location assignment for the kitting (order picking) process. The physical layout, the resource

fleet, and the milkrun schedule are held fixed. Replenishment scheduling, workforce planning, dynamic routing, and equipment investment are outside the scope, so that the effect of slotting alone can be isolated.

II. LITERATURE REVIEW

This section reviews work across four threads: discrete-event simulation (DES) in warehousing, warehouse layout optimization with multi-criteria classification, automated storage and retrieval systems (AS/RS), and decision support systems (DSS) in warehouse management. It closes by identifying the research gap that motivates the proposed line-aware slotting policy.

A. Simulation-Based Warehouse Analysis

Simulation is widely used to evaluate warehouse performance and capacity. Discrete-event simulation (DES) can represent stochastic arrivals, processing times, and shared-resource constraints. Negahban and Smith [18] show that simulation is effective for finding bottlenecks, assessing storage utilization, and testing alternative configurations in manufacturing and warehouse settings without disrupting real operations. Hapsari et al. [12] reach a similar conclusion for a pharmaceutical warehouse, where a simulation surfaced bottlenecks that static ABC analysis could not. A recurring limitation is that many studies focus on a single aspect, such as routing or capacity, whereas a real warehouse needs an integrated view that combines layout, routing, and operational decisions.

B. Warehouse Layout Optimization

Aradiansyah et al. [2] study picking inefficiencies caused by unoptimized layouts and irregular placement, and note that picking consumes much more time than checking, packing, or loading. They combine Double ABC analysis, which classifies inventory by both pick frequency and outer volume, with Arena-based DES to model outbound processes. A one-way ANOVA confirmed a significant reduction in moving time, showing that layout changes reduce operator travel distance. Their work provides a validated framework for optimizing limited-space warehouses, and it is the methodological basis for the Double ABC policy evaluated here.

C. Automated Storage and Retrieval Systems (AS/RS)

Automation technologies such as AS/RS are often proposed to improve warehouse performance. AS/RS use cranes or robotic systems in high-density racks, which increases space efficiency and reduces dependence on manual labour. The literature reports advantages including higher storage density from narrower aisles, shorter travel time, better accuracy, and safer operation through reduced human-vehicle interaction. These systems also carry trade-offs, mainly high investment cost and reduced flexibility [20].

AS/RS design involves two parts: physical design and control policies [20]. Physical design fixes rack dimensions, aisle configuration, and density; control policies set the storage-assignment strategy, such as dedicated, random, or class-based

storage. Class-based storage is widely preferred because it balances accessibility against space use. Performance is usually judged by travel time, throughput, waiting time, and resource utilization. AS/RS represents the highest level of physical automation, but the present study shows that meaningful gains are available from slotting policy alone, without the capital cost of full automation.

D. Decision Support Systems in Warehouse Management

Warehouse design, storage location assignment, and order picking are often studied separately, although in practice they are linked. Moghadam et al. [7] show that treating them independently leads to suboptimal performance, because layout choices dictate where items are stored and how efficiently they can be picked. The present study evaluates storage and classification policies without altering the physical layout, an approach supported by Amorim-Lopes et al. [1], who show that combining simulation with optimization can reduce travel distance and retrieval time even when the layout is unchanged.

Decision support systems (DSS) are increasingly used in warehouse management to support data-driven decisions. They help managers analyse operational data and evaluate alternative scenarios for layout, material flow, and resource use, replacing purely intuitive judgement. The literature indicates that DSS approaches improve process visibility and decision quality, but they tend to emphasize high-level decision frameworks rather than detailed modelling of the underlying operations. Combining the DSS philosophy with simulation gives a more complete solution, and that is the approach taken here: the operations are modelled in detail first, which produces a validated, data-driven basis that could later feed a broader decision-support framework.

E. Research Gap and Contribution

The literature covers storage assignment, simulation-based analysis, and layout improvement extensively, but most studies evaluate materials by aggregate demand and treat items as independent [10], [17]. In synchronized, production-driven warehouses, however, material flow is shaped by the requirements of specific production lines and by kitting: components are consumed in coordinated sets, so their relative locations directly affect kit preparation time [11], [19]. These dependencies are rarely addressed, and classification methods are usually applied uniformly without adapting to such operational characteristics [13], [20]. The gap is therefore not a shortage of assignment methods, but the failure to adapt and evaluate them in a real warehouse that accounts for line-based consumption and kit-level spatial dependencies.

The contribution of this study is a dependency-aware classification approach, the proposed *Line-aware Slotting* policy, and its evaluation against five alternatives within a discrete-event simulation framework. The study shows how established storage strategies can be adapted to real operating conditions rather than to aggregate demand, and it provides a statistically validated, application-oriented method [2], [15] that reduces

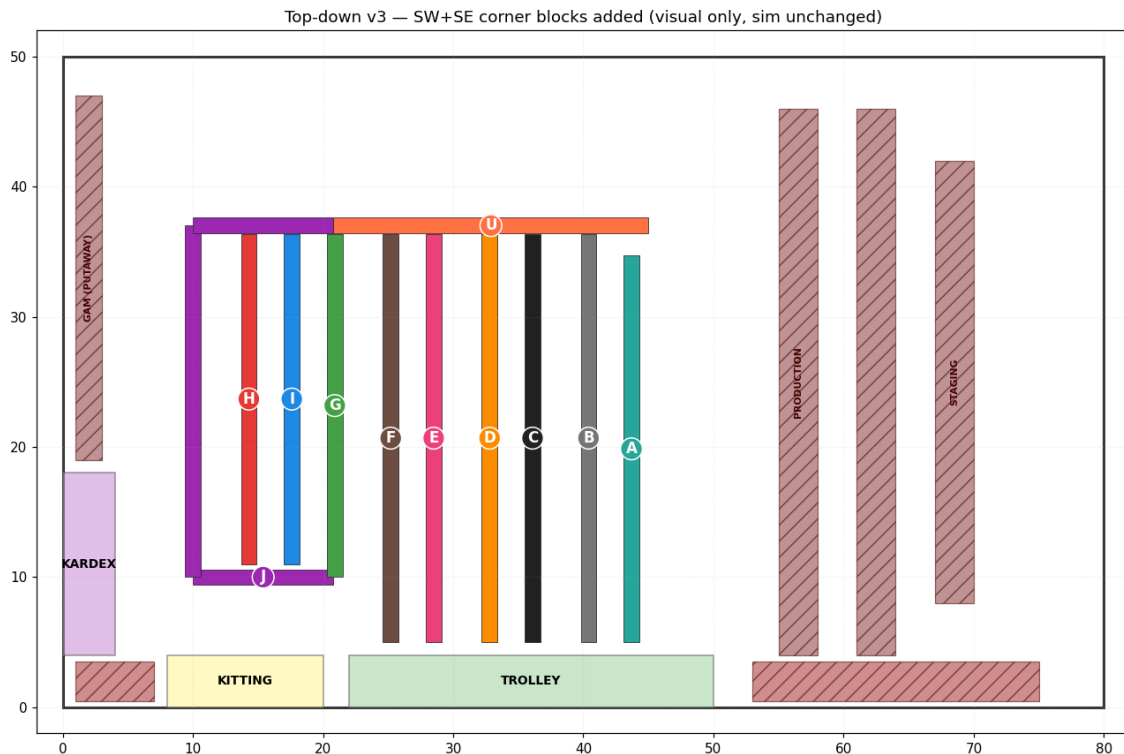


Fig. 1. Top-down layout of the modelled warehouse. Rack rows are labelled A through J and U; kit corridors (about 2 m) alternate with reach-truck aisles (about 3 m) along the east-to-west sequence.

waiting waste and improves material synchronization without physical layout changes.

III. METHODOLOGY

This section describes the framework used to analyse the current operation, develop alternative storage and classification policies, and evaluate their impact. The approach is built on historical SAP data, detailed system modelling, and discrete-event simulation. By replicating the warehouse in software, the policies are tested under identical, controlled conditions, which allows a fair comparison against fixed metrics. Figure 2 summarizes the five phases of the work.

A. Data Collection

The data came directly from the company’s SAP system, in cooperation with company representatives. The analysis uses operational data from 2 January to 18 May 2026, covering 40,804 kit orders across 103 active working days.

The primary dataset is the SAP ZWM92 transaction log, which records material withdrawals, transaction quantities, and the flows associated with kitting and production support. These records were combined with withdrawal quantities to determine the actual usage frequency of each component. During preprocessing, records with zero or negative withdrawal quantities were removed to keep the dataset clean.

Several supplementary datasets were added: current storage bins, per-line material usage, and the company’s existing

ABC/FMR results. An independent Pareto analysis was carried out on total consumption: materials up to 80% of cumulative consumption were classified as A, the next 15% (to 95%) as B, and the remaining 5% as C [8], [16], as defined in Equation (1). This classification was then compared against the existing one to document differences.

$$\text{Class}(i) = \begin{cases} A, & \text{if } P_i \leq 0.80, \\ B, & \text{if } 0.80 < P_i \leq 0.95, \\ C, & \text{if } P_i > 0.95, \end{cases} \quad P_i = \frac{\sum_{j=1}^i c_j}{\sum_{j=1}^n c_j}, \quad (1)$$

where c_j is the consumption of material j ranked in decreasing order and P_i is the cumulative consumption share up to rank i .

Technical layout drawings and rack documents supplied the physical dimensions: rack modules, storage levels, fast-mover shelves, aisle widths, and the relationships between storage and kitting zones. Bill of Materials (BOM) data and historical kitting records were used to map material relationships and kit-based usage [4].

Beyond the SAP data, operational parameters were collected through field observation. A video time-motion study recorded 2,319 micro-events on the F400 production line, capturing operator walking speeds, reach-truck movement and lifting times, manual picking durations, and the operator waiting time caused by reach-truck dependency. Operator allocation, fast-

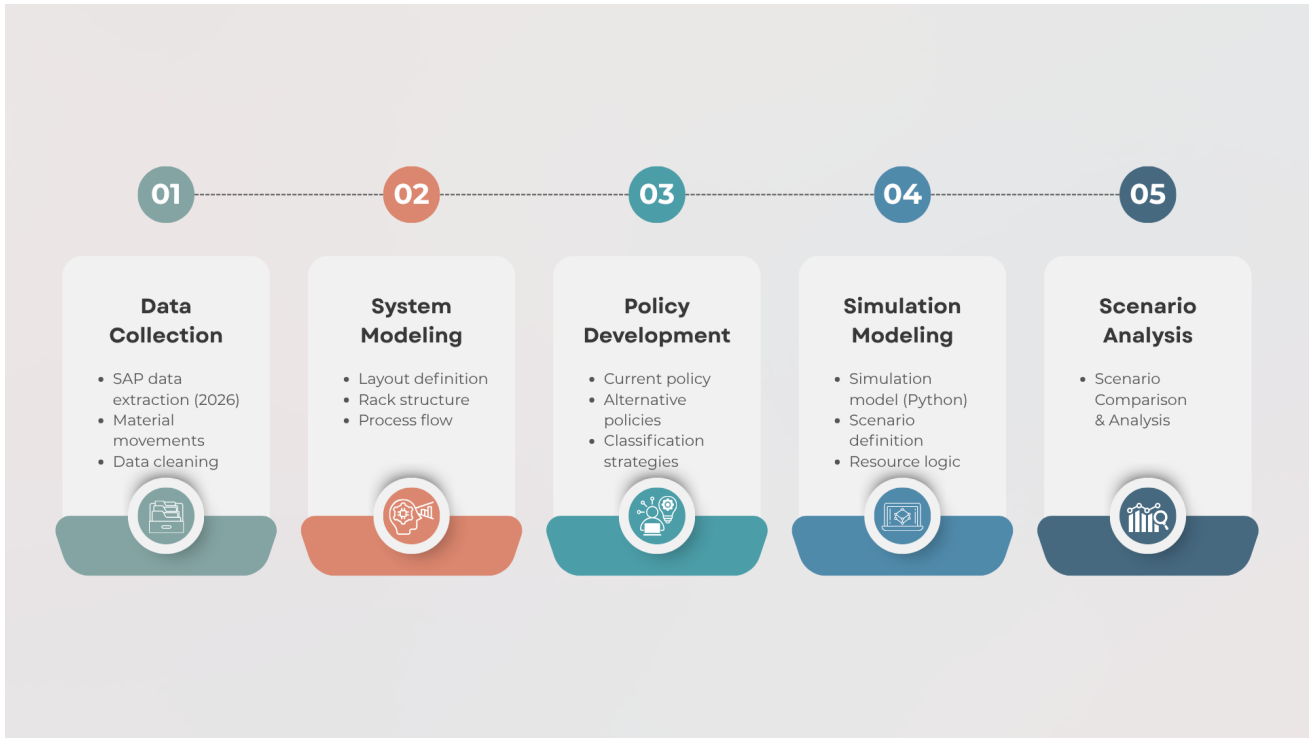


Fig. 2. The five-phase methodology framework for the simulation-based warehouse study.

mover usage, reachable levels, and handling patterns were recorded as well. Together these sources formed the quantitative basis for policy development, modelling, and evaluation.

B. System Modelling

The warehouse is represented within a discrete-event framework. Storage zones are modelled as distinct locations; operators and reach trucks (RTs) are modelled as active resources; the milkrun trains are transport entities; and production orders are kit-preparation requests triggered by demand.

The physical layout has 11 rack modules with a total capacity of 3,203 pallet positions. Materials are unitized on standard Euro pallets (80 × 120 cm). Reach-truck aisles measure 300 to 320 cm and dedicated kit corridors are 200 cm wide, as confirmed by the facility AutoCAD measurements. Rack heights reach about 7.5 m, which requires vertical storage handling [4].

A central design feature is the separation between standard storage and fast-mover sections. The fast-mover zone, the three lowest rack levels up to about 206 cm, is the manual band, where operators can reach materials without a reach truck. Here the racks allow decoupled work: operators pick from one side while replenishment is performed from the other, which removes direct contact between personnel and equipment. Standard and upper-level positions require reach-truck support for all retrievals [20].

The current state follows a hybrid logic in which assignments are influenced by line requirements, capacity heuristics, and the existing ABC/FMR classification. In practice,

operations deviate from these rules because of space limits and immediate replenishment needs, which fragments placement across zones. This fragmentation adds travel distance, increases reach-truck dependency, and creates waiting time during kit preparation, especially for materials in upper levels. The model captures these interactions when assessing the alternative policies [10].

C. Policy Design

This section describes the storage and classification policies evaluated in the study. The policies use different elements of the operational data: physical accessibility, usage frequency, retrieval time, and production-line and kit dependencies.

1) *Baseline and Capacity Heuristic (Control Group)*: Two reference policies form the control group:

- **Baseline (Actual SAP)**. This policy reproduces the historical placement of the warehouse, reflecting patterns accumulated over years of operation and daily replenishment. It is the main reference point for measuring the real-world impact of the alternatives [15].
- **Capacity heuristic**. Materials are assigned to the first available slot that meets the item's storage requirement, with no line-aware logic or demand dependency. Comparing the proposed policy against this heuristic isolates the gain from dependency-aware slotting versus simple space filling [20].

2) *Double ABC Analysis Policy*: The Double ABC policy follows the method of Ardiansyah et al. [2], which shows that combining two classification dimensions gives a more gran-

ular prioritization than a single-axis ABC. Because reliable material-volume data were not available, the second dimension uses material cost rather than volume. The first dimension captures economic importance (cost) and the second captures operational importance (frequency). The two criteria are combined in Table I.

The intersection of the two criteria gives a 3×3 matrix with nine categories (AA through CC):

- **AA (high cost, high frequency):** highest priority, assigned to prime locations such as fast-mover shelves and lower levels.
- **CC (low cost, low frequency):** lowest priority, assigned to upper levels or less accessible locations.
- **Intermediate (AB, BA, and so on):** placed by their combined cost and frequency, balancing accessibility against inventory value.

This moves the plant beyond simple frequency-based classification and prevents rack capacity from being consumed by bulky, low-demand items.

3) *Time-Based Classification Policy:* Unlike static methods based on frequency or volume, the time-based policy uses the accessibility of each location. In a high-bay warehouse, total retrieval time is dominated not by horizontal travel but by vertical lift and the need for reach-truck intervention [4], [14]. The policy estimates an access time for each location from travel speed, lift speed, and reach-truck handling, then assigns frequently used materials to the locations with the shortest access time. Table II summarizes the mapping. By using time rather than distance alone, the policy aims to reduce lead time and waiting and to support line synchronization [19], [17].

4) *Line-aware Slotting (Proposed Policy):* The proposed *Line-aware Slotting* policy is the core contribution of this study. It shifts the focus from warehouse-wide averages to production-line dependencies. The implementation has two stages. First, an independent ABC analysis is run for each production line rather than for the warehouse as a whole, so materials are ranked by their consumption within each line, with A-class defined as the top 80% of cumulative consumption. This removes the distortion caused by aggregated data and reflects the true criticality of each material for its line. Second, the policy uses historical kitting and BOM data to identify items that are frequently picked together and places them in the same rack module or aisle. Materials shared across several lines are positioned centrally in high-accessibility zones to reduce unnecessary travel [11], [17].

The placement logic prioritizes the manual band, the three lowest levels that operators can reach by hand. By keeping high-frequency, line-specific materials in this zone, the policy achieves two goals at once: proximity, so materials sit close to the line they serve, and reach-truck decoupling, so frequent picks no longer require an RT [5]. This directly attacks the waiting waste in the kit corridors and reduced reach-truck utilization to 1.3% in the simulation, as reported in Section IV.

D. Simulation Model

The model was built with the Python SimPy library to represent the interaction of warehouse operations and to evaluate the alternative policies. Simulation was chosen because it can capture resource congestion and stochastic arrivals under realistic conditions without disrupting the real operation [10], [18], [3]. The implementation is organized into 17 source modules that handle the full lifecycle: data loading and SAP ZWM92 bin decoding, warehouse geometry, rectilinear routing with rack-body obstacles, slotting policy management, the SimPy process logic, KPI recording, and validation through sensitivity sweeps. Figure 3 shows the lifecycle of one kit order in the model.

Order arrivals are generated from the demand pattern in the SAP ZWM92 log, using empirical inter-batch gaps and batch sizes. The inter-batch arrival gap has a mean of 5.36 minutes and a coefficient of variation of 2.04, while the mean batch size is 4.68 kits. The coefficient of variation of the gap is well above 1.0, which rules out an exponential model and justifies sampling from the empirical distribution rather than a fitted theoretical one [16], [22]. Service times use lognormal distributions parameterized from the F400 time-motion study, with a mean operator pick time of 0.113 minutes, a reach-truck pick-and-place time of 0.110 minutes, a manual high-level pick penalty of 0.102 minutes, and a Kardex retrieval time of 0.113 minutes. The coefficient of variation is defined in Equation (2), and each lognormal distribution is moment-matched to its measured mean and coefficient of variation through Equations (3) and (4).

$$CV = \frac{s}{\bar{x}} \quad (2)$$

$$\sigma = \sqrt{\ln(1 + CV^2)} \quad (3)$$

$$\mu_{\log} = \ln(\bar{x}) - \frac{1}{2}\sigma^2 \quad (4)$$

where $\bar{x}$ and s are the sample mean and standard deviation of the measured time, and σ and $\mu_{\log}$ are the shape and location parameters of the lognormal distribution; the location term in Equation (4) keeps the distribution mean equal to $\bar{x}$ for any σ .

When an order is triggered, the model looks up the storage location of each requested material, computes the shortest obstacle-free rectilinear travel path for the assigned resource, and performs the pick. Travel distance uses the rectilinear (Manhattan) metric of Equation (5), evaluated between pick points and summed over the route; the router searches a set of aisle waypoints and returns the shortest path that avoids rack bodies. If a required resource is busy, the order waits in a queue and the waiting time is recorded.

$$d(A, B) = |x_A - x_B| + |y_A - y_B| \quad (5)$$

The architecture lets the storage-assignment logic be replaced without changing the rest of the model, so every policy faces the same demand stream and resource pool, which keeps

TABLE I
DOUBLE ABC CLASSIFICATION MATRIX.

Dimension	Classes	Thresholds
Cost (C)	High / Medium / Low	by unit cost
Frequency (F)	High / Medium / Low	> 80% / 80–95% / 95–100% cumulative

TABLE II
TIME-BASED POLICY: STORAGE ZONE ASSIGNMENT BY ACCESS TIME.

Relative access time	Storage assignment	Access method
Minimal	Fast-mover shelves	Manual picking
Short	Lower rack levels	Reach truck, minimal vertical travel
Moderate	Mid rack levels	Reach truck, standard vertical travel
Extended	Upper levels	Reach truck, significant vertical travel

the comparison fair. To check spatial accuracy, a 3D viewer was built in Three.js that renders the layout from the AutoCAD DWG file and the eleven rack PDF drawings. Visual inspection of the exported movement paths confirmed that agents follow the corridors and do not pass through rack bodies [2], [21].

E. Performance Metrics

The policies are evaluated on five key performance indicators (KPIs):

- **Kitting lead time (min/order)**. Time from order creation until the completed kit is ready at the staging area for milkrun pickup. Lower lead time means faster supply to the lines.
- **Operator waiting time (min/order)**. Time the kitting operator spends idle, waiting for reach-truck support to retrieve upper-level materials.
- **Reach-truck utilization (%)**. Share of total reach-truck capacity actively used for retrieval, defined in Equation (6).
- **Throughput (orders/day)**. Number of kits completed per working day, defined in Equation (7).
- **Operator walking distance (m/order)**. Total rectilinear path length an operator walks to complete one kit. This metric checks that any lead time reduction comes from smarter placement, not from making the operator walk further.

$$U_{RT} = \frac{\sum_{r=1}^{N_{RT}} t_r^{busy}}{T_{sim} N_{RT}} \quad (6)$$

$$\Lambda = \frac{N_{completed}}{T_{sim}} \quad (7)$$

where t_r^{busy} is the busy time of reach truck r , N_{RT} is the number of reach trucks, T_{sim} is the simulated horizon, and $N_{completed}$ is the number of completed kit orders.

F. Verification and Validation

1) *Verification*: Verification checked that the model was built correctly and behaves as intended, through code-level checks and logical-flow tests. At the code level, tests covered

the SAP bin decoder, the Kardex routing logic, and resource tracking. A ten-step control list confirms that the modelled pallet capacity is consistent with the layout file before any result is recorded, that every material has a valid location, and that pick counts are correct. The model checks that its 3,101 modelled positions reconcile with the 3,203 pallet positions stated in the rack drawings (the difference reflects narrower feeder bays and a cart-parking slot) before a run proceeds. Fixed random seeds make each run exactly reproducible.

The logical flow was then tested step by step, covering order creation, picking, and reach-truck use. The model was also run on small scenarios whose expected travel and waiting times could be computed by hand and compared against the output. Finally, extreme scenarios were tested: with all materials on the lowest levels the model produced zero reach-truck waiting, and with all materials on the top levels it produced a clear rise in waiting and reach-truck use. As Table III shows, the difference between the hand calculation and the simulation was below 1% in every scenario.

To keep the comparison between policies fair, common random numbers (CRN) were used, so every policy faces the same sequence of orders and any difference comes from the storage strategy rather than from random variation.

2) *Validation*: Validation assessed whether the model represents the real warehouse, following the Sargent framework [21] with both operational review and statistical checks. First, the physical and operational elements, including the layout, racks, fast-mover areas, and milkrun cycles, were reproduced from SAP data, drawings, BOM relationships, and field observation, and the process logic was reviewed with company representatives.

Second, a chi-square goodness-of-fit test compared the simulated pick distribution against the historical SAP ZWM92 data. The pooled test gave $\chi^2(7) = 535.05$ ($p < 0.001$), pooled from the ten-rack restricted test $\chi^2(9) = 537.68$ to satisfy Cochran's rule, with a Cramér's V of 0.22, which indicates a small-to-medium effect. The statistic and effect size are defined in Equations (8) and (9). Because the simulation assigns pick locations using its own logic rather than replaying history, an exact bin-for-bin match is neither expected nor

How the simulation handles one kit order

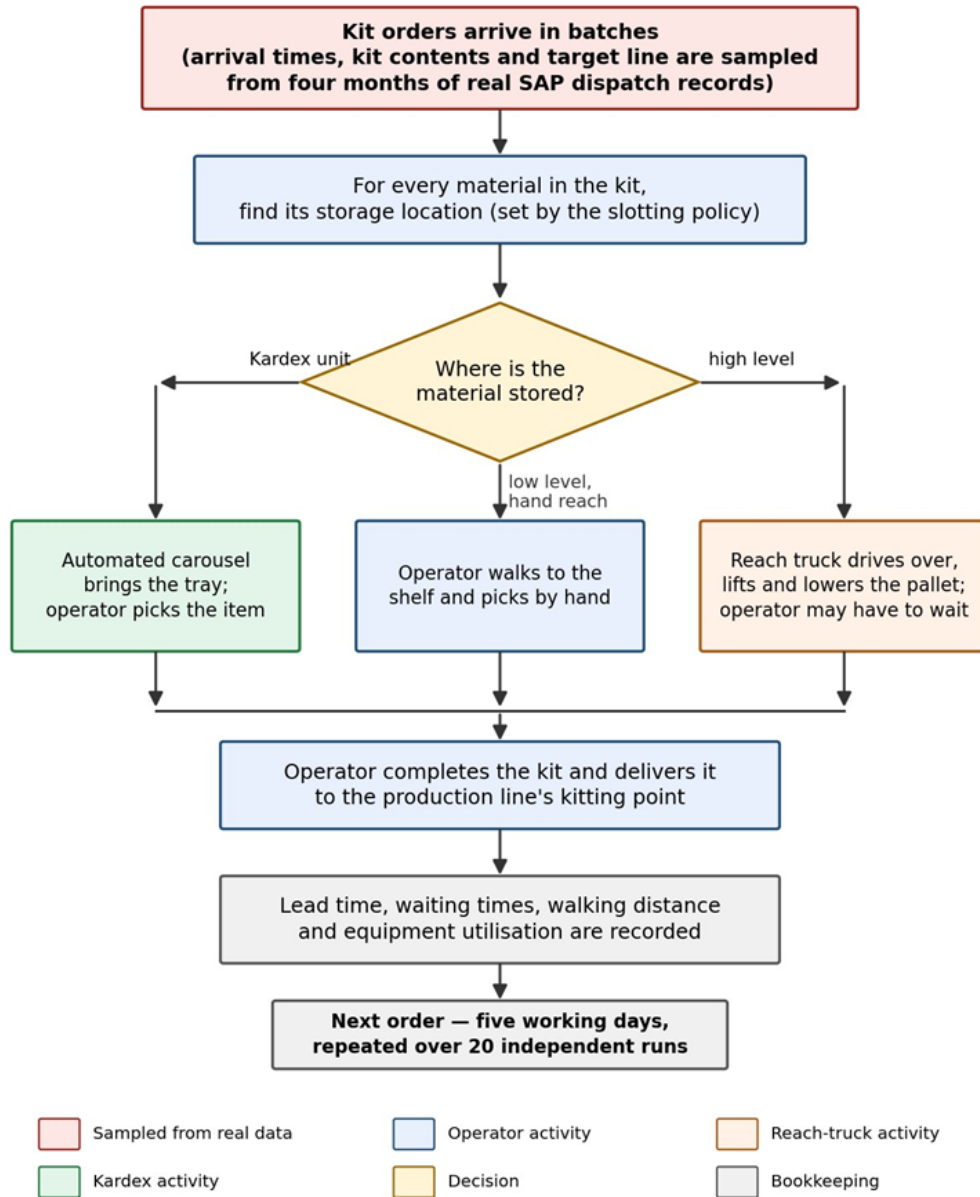


Fig. 3. Lifecycle of a single kit order in the simulation.

TABLE III
VERIFICATION TEST OUTCOMES.

Test scenario	Expected behaviour	Outcome
Single lower-rack retrieval	Minimal wait, no RT needed	Pass
Single upper-rack retrieval	RT request generated	Pass
No upper-rack materials	Near-zero waiting time	Pass (0.12 min)
All upper-rack materials	High RT use, long wait	Pass

desired; the rack ranking, however, agrees, and the busiest racks in the log (E, G, and D) are also the busiest in the model. Table IV summarizes the checks.

$$\chi^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i} \quad (8)$$

$$V = \sqrt{\frac{\chi^2}{n(k-1)}} \quad (9)$$

where O_i and E_i are the observed and expected pick counts in rack i , k is the number of racks, and n is the total number of picks.

The daily volume calibration matched the real average of about 396 kits per day within 1.1%. The routing model was checked on simulated agent paths, with no agent passing through a rack body. Reproducibility checks confirmed stable results across seeds. Together these checks support the use of the model for comparing storage policies.

G. Experimental Design and Statistical Analysis

Each policy was run for $N = 20$ independent replications. Common random numbers were used across policies: replication i of every policy uses seed $42 + i$ for the arrival stream, so all policies face the same demand trajectory in a given replication. This pairing removes demand-side variance from the policy comparison. Within a policy the replications use distinct seeds and are therefore independent; across policies, the replications are correlated by design through the shared seeds, which is exactly what the paired analysis exploits. A warm-up period was excluded from each run to reduce initialization bias, so only post-warm-up output was retained.

For each replication, the model recorded the mean of each KPI. The sample mean, standard deviation, and a 95% confidence interval were then computed for each policy and KPI. Because $N = 20$, the interval uses the t -distribution with 19 degrees of freedom ($t_{0.025,19} = 2.093$), as in Equation (10).

$$\bar{x} \pm t_{0.025,19} \frac{s}{\sqrt{N}} \quad (10)$$

A one-way ANOVA was used as the primary test, with storage policy as the factor (six levels) and the per-replication KPI as the response, at a significance level of $\alpha = 0.05$. Because each KPI is an average of many events within a replication, its distribution across replications is approximately normal, and the sample sizes are equal, so the assumptions for the parametric test are reasonable. The ANOVA statistic is given in Equation (11) and the hypotheses in Equation (12).

$$F = \frac{MS_B}{MS_W} = \frac{SS_B/(k-1)}{SS_W/(N_{\text{tot}} - k)} \quad (11)$$

$$H_0 : \mu_1 = \mu_2 = \dots = \mu_6 \quad H_1 : \exists i \neq j \text{ such that } \mu_i \neq \mu_j \quad (12)$$

Where the ANOVA was significant, Tukey's Honestly Significant Difference (HSD) test identified which pairs differ.

In addition, common-random-number paired t -tests compared each alternative directly against the Actual SAP baseline. The paired test (Equation (13)) uses the per-replication differences, which removes the shared demand variance and gives a tighter contrast.

$$t = \frac{\bar{d}}{s_d/\sqrt{N}}, \quad d_i = \text{KPI}_{\text{alt},i} - \text{KPI}_{\text{base},i} \quad (13)$$

with hypotheses $H_0 : \mu_d = 0$ and $H_1 : \mu_d \neq 0$. A negative mean difference favours the alternative policy. The KPI values were exported per replication for the analysis and for cross-checking in Minitab.

IV. RESULTS AND DISCUSSION

This section reports the simulation results for the six policies. Section IV-A gives the replication-level KPIs and confidence intervals, Section IV-B the sensitivity analysis, Sections IV-C and IV-D the statistical tests, Section IV-E the effect size, and Section IV-F the discussion.

A. Simulation Results and Policy Comparison

The outputs are stable across the 20 replications. The proposed *Line-aware Slotting* policy reached a mean kitting lead time of 6.91 minutes with a narrow 95% confidence interval of [6.44, 7.38]. Its mean operator waiting time was 4.26 minutes, and reach-truck utilization was only 1.27%, which is consistent with placing frequently used materials in hand-reachable positions. Throughput stayed at about 398.6 orders per day, so the policy improves preparation without reducing output. Tables V through VIII report lead time, waiting time, reach-truck utilization, and throughput; Table IX reports walking distance. Figure 4 compares the policies across four panels.

The walking-distance results in Table IX are central to the interpretation. *Line-aware Slotting* walks 92.4 m per order, slightly more than the 89.0 m of the current SAP placement, yet its lead time is about half. The gain therefore does not come from shorter walking; it comes from removing reach-truck calls from the kit path, which is examined in Section IV-F.

B. Sensitivity Analysis

A one-at-a-time (OAT) sensitivity analysis was run on nine timing and operational parameters, using the capacity heuristic as the reference policy ($N = 5$ replications per cell, baseline lead time 14.00 minutes). Each parameter was varied by $\pm 20\%$ from its calibrated value while the others were held constant. Table X reports the resulting minimum and maximum lead time and the swing, defined as the range divided by the baseline.

Lead time is most sensitive to operator walking speed (38.3% swing) and to the arrival rate (37.4%), followed by the reach-truck lift time per level (18.5%). The remaining parameters move lead time by less than 7%. The conclusions are therefore robust to estimation error in the smaller timing constants. Figure 5 shows the same result as a tornado chart.

TABLE IV
VALIDATION SUMMARY (SARGENT FRAMEWORK).

Validation check	Result	Key metric
Distributional fit (chi-square)	Pass (disclosed)	$\chi^2(7) = 535.05$, $V = 0.22$
Daily volume calibration	Pass	Error 1.1% vs ≈ 396 kits/day
Route model (zero crossings)	Pass	No agent crosses a rack body
Reproducibility	Pass	Stable across seeds

TABLE V
KITTING LEAD TIME BY POLICY ($N = 20$ REPLICATIONS).

Policy	Mean (min)	Std. dev.	95% CI
Line-aware Slotting	6.91	0.99	[6.44, 7.38]
Baseline (Actual SAP)	12.37	2.29	[11.30, 13.44]
Baseline (Heuristic)	14.71	2.84	[13.38, 16.04]
Double ABC	14.74	2.89	[13.38, 16.09]
Usage-based ABC	15.24	3.08	[13.80, 16.68]
Travel-distance Optimized	24.60	7.19	[21.23, 27.96]

TABLE VI
OPERATOR WAITING TIME BY POLICY ($N = 20$ REPLICATIONS).

Policy	Mean (min)	Std. dev.	95% CI
Line-aware Slotting	4.26	0.98	[3.80, 4.72]
Baseline (Actual SAP)	8.45	2.29	[7.38, 9.52]
Baseline (Heuristic)	10.40	2.85	[9.07, 11.73]
Double ABC	10.40	2.90	[9.04, 11.76]
Usage-based ABC	10.81	3.10	[9.37, 12.26]
Travel-distance Optimized	19.03	7.20	[15.66, 22.40]

TABLE VII
REACH-TRUCK UTILIZATION BY POLICY ($N = 20$ REPLICATIONS).

Policy	Mean (%)	Std. dev.	95% CI
Line-aware Slotting	1.27	0.28	[1.14, 1.40]
Baseline (Actual SAP)	21.71	2.46	[20.55, 22.86]
Baseline (Heuristic)	22.82	2.71	[21.55, 24.09]
Double ABC	24.58	2.83	[23.25, 25.90]
Usage-based ABC	24.41	2.74	[23.12, 25.68]
Travel-distance Optimized	43.95	4.72	[41.74, 46.16]

TABLE VIII
THROUGHPUT BY POLICY ($N = 20$ REPLICATIONS).

Policy	Mean (orders/day)	Std. dev.	95% CI
Line-aware Slotting	398.58	44.49	[377.76, 419.40]
Baseline (Actual SAP)	398.51	44.45	[377.71, 419.31]
Baseline (Heuristic)	398.38	44.31	[377.64, 419.12]
Double ABC	398.39	44.29	[377.66, 419.12]
Usage-based ABC	398.42	44.42	[377.63, 419.21]
Travel-distance Optimized	396.17	43.16	[375.98, 416.37]

TABLE IX
OPERATOR WALKING DISTANCE BY POLICY ($N = 20$ REPLICATIONS).

Policy	Mean (m/order)	Std. dev.	95% CI
Baseline (Actual SAP)	89.0	1.30	[88.4, 89.6]
Line-aware Slotting	92.4	1.79	[91.6, 93.3]
Travel-distance Optimized	96.1	1.65	[95.3, 96.9]
Double ABC	104.6	1.70	[103.8, 105.4]
Baseline (Heuristic)	106.1	1.90	[105.2, 107.0]
Usage-based ABC	108.8	1.67	[108.0, 109.6]

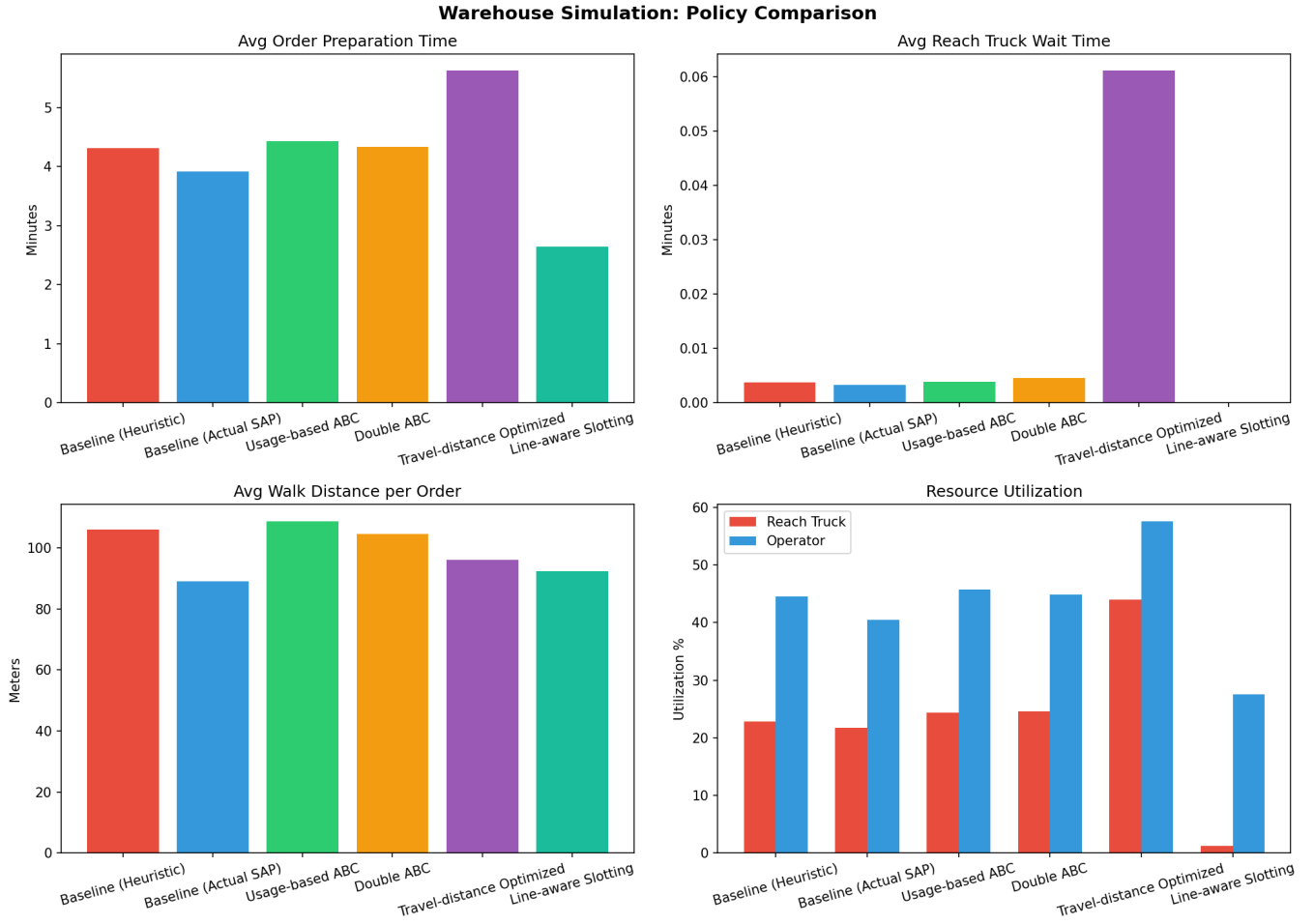


Fig. 4. Policy comparison across four panels: average order preparation time, average reach-truck wait time, average walking distance per order, and resource utilization (reach truck and operator). All values are means over $N = 20$ replications with common random numbers.

TABLE X
ONE-AT-A-TIME SENSITIVITY OF KITTING LEAD TIME TO $\pm 20\%$ PARAMETER CHANGES (CAPACITY HEURISTIC REFERENCE, BASELINE 14.00 MIN; ★ MARKS A SWING ABOVE 15%).

Parameter	Base value	Min lead (min)	Max lead (min)	Swing
Operator walking speed	50 m/min	12.03	17.40	38.3% ★
Order inter-arrival time	5.36 min	12.29	17.53	37.4% ★
Reach-truck lift time per level	0.25 min	12.74	15.33	18.5% ★
Reach-truck travel speed	100 m/min	13.65	14.58	6.6%
Kardex carousel time	0.40 min	13.74	14.27	3.8%
Operator pick time	0.113 min	13.79	14.21	3.0%
Kardex pick time	0.113 min	13.82	14.19	2.6%
Reach-truck pick/place time	0.110 min	13.91	14.11	1.5%
Manual high-level pick penalty	0.102 min	13.99	14.06	0.5%

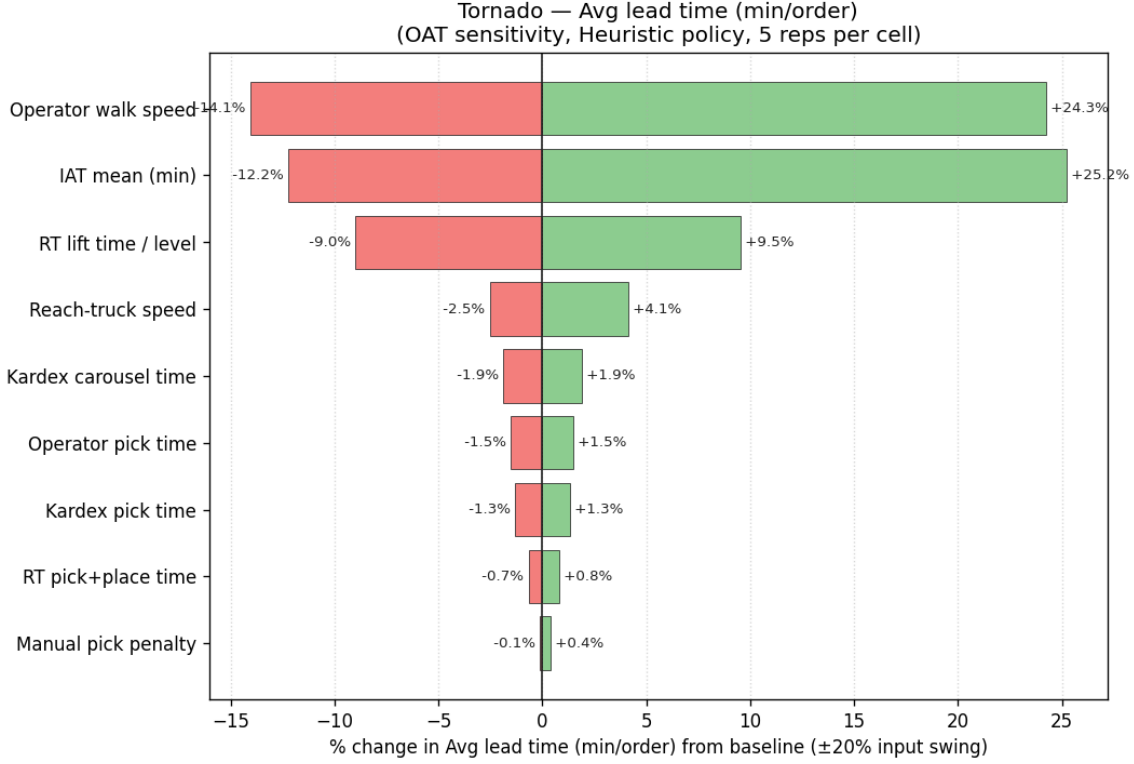


Fig. 5. Tornado chart of the one-at-a-time sensitivity of mean kitting lead time to $\pm 20\%$ changes in each parameter.

A stress test was also run by doubling the arrival rate (twice the system load). Under this heavier load the lead-time advantage of *Line-aware Slotting* over the SAP baseline grows from 5.46 minutes at normal load to 20.7 minutes, which shows that the policy becomes more valuable as the system is pushed harder.

C. Statistical Hypothesis Testing

The one-way ANOVA (Table XI) shows that the choice of storage policy has a significant effect on kitting lead time, waiting time, reach-truck utilization, and walking distance ($p < 0.001$ in each case). Throughput does not differ between policies ($F = 0.01$, $p \approx 1.0$), which confirms that the system is not order-constrained: it completes about the same number of kits per day regardless of policy, so the gains appear as lower lead time and waiting rather than as higher output.

D. Paired t -Test Results and Policy Contrast

Table XII reports the common-random-number paired t -tests on kitting lead time against the current SAP placement. *Line-aware Slotting* reduces lead time by 5.46 minutes per order ($t = -17.47$, $p = 3.68 \times 10^{-13}$), and the interval does not approach zero. Every other alternative is significantly slower than the current state: the capacity heuristic and the two ABC variants add roughly 2.3 to 2.9 minutes per order, and the travel-distance policy adds 12.23 minutes.

The pattern is clear: frequency-based or distance-based slotting alone is not enough for this warehouse. The drivers

of improvement are line-based accessibility and a reduced dependence on reach trucks. The *Line-aware Slotting* policy is the only alternative that improves on the current state, because it considers both the production-line relationship and the manual accessibility of materials, which lowers reach-truck intervention and stabilizes the kitting process.

E. Practical Effect Size: Cohen's d

ANOVA and the paired t -tests establish statistical significance; Cohen's d measures the practical size of the improvement. For the primary contrast, *Line-aware Slotting* versus the SAP baseline, the effect size is $d = 3.10$ (mean difference 5.46 min, pooled standard deviation 1.76 min), as in Equation (14) and Table XIII. Since $d \geq 0.8$ is conventionally large, a value of 3.10 indicates a very large practical effect, which confirms that the improvement is operationally meaningful and not just statistically detectable.

$$d = \frac{\bar{x}_1 - \bar{x}_2}{s_p}, \quad s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} \quad (14)$$

F. Discussion

Storage-assignment research has repeatedly found that good slotting can cut order-picking travel by 10 to 30% relative to random or class-based baselines. The present result is consistent in magnitude, since *Line-aware Slotting* walks 92.4 m

TABLE XI
ONE-WAY ANOVA ACROSS THE SIX POLICIES FOR EACH KPI ($N = 20$ PER POLICY). H_0 : ALL POLICY MEANS ARE EQUAL.

KPI	F	d.f.	p -value	Decision at $\alpha = 0.05$
Kitting lead time	47.07	(5, 114)	3.55×10^{-26}	Reject H_0
Total waiting time	33.09	(5, 114)	9.86×10^{-21}	Reject H_0
Reach-truck utilization	430.06	(5, 114)	3.19×10^{-72}	Reject H_0
Walking distance	464.15	(5, 114)	5.12×10^{-74}	Reject H_0
Throughput	0.01	(5, 114)	1.000	Fail to reject H_0

TABLE XII
REPLICATION-PAIRED t -TESTS ON KITTING LEAD TIME AGAINST THE CURRENT SAP PLACEMENT (COMMON RANDOM NUMBERS, $N = 20$ PAIRS). A NEGATIVE DIFFERENCE FAVOURS THE ALTERNATIVE POLICY.

Policy vs. current placement	Mean diff. (min)	t	p -value	Significant?
Capacity heuristic	+2.34	11.27	7.47×10^{-10}	Yes
Usage-based ABC	+2.87	9.97	5.50×10^{-9}	Yes
Double ABC	+2.37	10.05	4.85×10^{-9}	Yes
Travel-distance Optimized	+12.23	10.06	4.82×10^{-9}	Yes
Line-aware Slotting	-5.46	-17.47	3.68×10^{-13}	Yes

TABLE XIII
COHEN'S d FOR THE PRIMARY POLICY CONTRAST (BENCHMARKS: $d \geq 0.2$ SMALL, ≥ 0.5 MEDIUM, ≥ 0.8 LARGE).

Contrast	Mean diff. (min)	Pooled SD (min)	Cohen's d
Line-aware vs. SAP baseline (lead time)	5.46	1.76	3.10

per order against 106.1 m for the capacity heuristic, a 13% reduction. It also adds a dimension that the layout literature rarely emphasizes: in a kitting environment with shared reach trucks, the dominant benefit of slotting is the reduction of resource contention rather than of travel distance.

The kitting context is central. A kit order bundles materials from across the warehouse into one set. Limere et al. [17] show that the choice between kitting and line stocking turns on handling cost and distance; this study shows that when reach trucks are shared, the handling cost is mainly queueing time. A policy that ignores this turns a resource that is idle about 78% of the time under SAP placement into one that is saturated at 44% under a pure travel-distance policy. Simulation is well suited to expose this non-linear behaviour, as also noted by Negahban and Smith [18] and Amorim-Lopes et al. [1].

Three features drive the advantage of *Line-aware Slotting*. First, the line-origin distance metric places each material near the line it serves most; because about 77% of kit demand originates at line-specific corridor points, this concentrates picks in small neighbourhoods. Second, the preference for no-reach-truck levels keeps materials at hand height and reserves the fleet for genuine high-level retrieval. Third, by spreading materials across several corridor segments, the policy avoids the local congestion that cripples the travel-distance strategy.

The Usage-based and Double ABC policies both follow the textbook rule to place fast movers nearest the output point. In a single-picker warehouse this is sound, but here there is no single dispatch dock: each line has its own kitting corridor. Collapsing that structure into one distance metric creates the same congestion as the travel-distance policy, only

to a smaller degree. The travel-distance policy is the extreme case, optimizing the wrong objective and producing a lead time about double the current state.

For the Manisa plant the implication is direct: re-slot fast movers line by line, with priority to the levels reachable by hand. At about 400 kit orders per day, the model estimates a mean reduction of 5.46 minutes per order, which is a substantial capacity gain over a shift with no new equipment or headcount. One caveat is that the 1.3% reach-truck utilization reported for the proposed policy is a picking-only figure. Putaway and replenishment, which would rise after moving high-turnover items to lower levels, are outside the model; quantifying that trade-off is a priority for future work. The comparison itself is fair, since every policy faces the same demand trajectory under common random numbers, the same resource pool, and the same modelling assumptions.

V. CONCLUSIONS AND FUTURE WORK

A. Conclusions

This study built a discrete-event simulation model of the Schneider Electric Manisa warehouse from historical SAP data, layout information, BOM relationships, and operational time studies, and used it to evaluate alternative storage-assignment policies under realistic conditions without disrupting the live operation.

Storage assignment has a clear effect on performance. Among the alternatives, the proposed *Line-aware Slotting* policy performed best: it reduced mean kitting lead time from 12.37 to 6.91 minutes per order, about 44%, and lowered reach-truck utilization from 21.7% to 1.3%. One-way ANOVA

and paired t -tests confirmed that these improvements are statistically significant and not the product of simulation noise.

The results show that the main source of inefficiency is not travel distance but the dependency on reach-truck intervention during kit preparation. By placing frequently used line-specific materials in hand-reachable positions, the proposed policy reduces waiting, improves synchronization between the warehouse and the production lines, and raises process efficiency. The broader message is that a substantial gain is available from data-driven storage assignment alone, without costly layout changes or automation.

B. Managerial Implications

The results offer practical guidance for managers who want to improve efficiency with limited investment. *Line-aware Slotting* can be introduced gradually, by first relocating the most frequently used materials to hand-reachable levels near the lines they serve. For a facility processing about 400 kit orders per day, the lead-time reduction is a real opportunity to cut waiting waste, improve resource use, and stabilize the process. The simulation framework itself is a decision-support tool: alternative policies can be evaluated virtually before any physical change, which lowers both risk and cost.

C. Limitations

Several modelling choices and scope limits should be kept in mind when reading the results.

Extrapolation of operational times. The timing parameters come from a video time-motion study of 2,319 micro-events on the F400 line and were then applied to the other eight product families. If F400 is not representative of slower or faster lines, the lead-time estimates will shift. The sensitivity analysis bounds this risk, but line-specific studies would refine the model.

Validation data. The model was fitted and validated against the same SAP ZWM92 dataset, because complete historical KPI records were not available for out-of-sample testing. The goodness-of-fit and daily-volume checks are therefore internal-consistency tests rather than true holdout validation.

SAP decoding and multi-bin splits. Only 750 active materials had fully decoded rack-bay-position codes, and 386 materials have multiple bins that the model does not split between forward and reserve. The Actual SAP baseline is therefore best read as an accurate SAP-plus-heuristic hybrid rather than a pure bin-for-bin replay of the layout.

Operator congestion. Operator congestion in shared aisles is not modelled; agents pass through each other freely. The model may therefore slightly underestimate lead time under high-load policies such as the travel-distance policy.

Scope limited to order picking. The scope is order picking and kitting only. *Line-aware Slotting* concentrates fast movers at low levels, which will increase putaway and replenishment frequency for those positions. The 1.3% reach-truck utilization is a picking-only figure, so the total reach-truck workload in a live deployment would be higher. Replenishment scheduling,

workforce planning, dynamic routing, and equipment allocation were deliberately left out so that the effect of storage assignment could be isolated.

D. Future Work

Several extensions are possible. Additional time studies on the other production lines would provide line-specific parameters and improve accuracy. Kit-based co-location could be combined with *Line-aware Slotting* to further reduce picking travel within a single order. A pilot in one warehouse zone would allow a before-and-after comparison against real performance. Finally, simulation could be combined with optimization methods such as mathematical programming or metaheuristics to generate and evaluate storage assignments automatically.

E. Sustainable Development

This study supports three of the United Nations Sustainable Development Goals [23]. SDG 9 (Industry, Innovation, and Infrastructure) is the most direct fit: replacing intuition-based assignment with a data-driven simulation is evidence-based industrial practice at essentially no capital cost, with no new equipment or expansion. SDG 12 (Responsible Consumption and Production) applies because the proposed slotting makes better use of the storage and handling equipment that already exists; lower reach-truck use means less energy per kit, and avoiding building expansion conserves land and materials. SDG 8 (Decent Work and Economic Growth) applies on the operator side: moving high-frequency items to the manual band reduces the ergonomic strain of mast-elevated retrieval and removes idle time spent waiting for a truck. These are modest, honest improvements that follow directly from the model's findings.

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APPENDIX

Table XV reports the mean of every KPI over the $N = 20$ replications for all six policies, as a single reference.

TABLE XIV
ABBREVIATIONS USED IN THE REPORT.

Abbreviation	Meaning
ABC	Activity-based classification (A/B/C demand classes)
ANOVA	Analysis of variance
BOM	Bill of materials
CRN	Common random numbers
CV	Coefficient of variation
DES	Discrete-event simulation
EPT	Electric pallet truck
FMR	Fast/medium/rare-mover classification
IAT	Inter-arrival time
KPI	Key performance indicator
OAT	One-at-a-time (sensitivity)
RT	Reach truck
SAP	Enterprise resource planning system of the plant
ZWM92	SAP warehouse-management dispatch transaction log

TABLE XV
FULL KPI TABLE, MEANS OVER $N = 20$ REPLICATIONS.

KPI	Current SAP	Capacity heuristic	Usage ABC	Double ABC	Travel-dist. opt.	Line-aware slotting
Lead time (min)	12.37	14.71	15.24	14.74	24.60	6.91
Prep time (min)	3.92	4.31	4.43	4.34	5.62	2.65
Total wait (min)	8.45	10.40	10.82	10.40	19.03	4.26
P95 lead time (min)	35.9	43.0	42.9	41.9	70.6	18.8
Walk distance (m)	89.0	106.1	108.8	104.6	96.1	92.4
Throughput (orders/day)	398.5	398.4	398.4	398.4	396.2	398.6
RT utilization	0.217	0.228	0.244	0.246	0.440	0.013
Operator utilization	0.405	0.446	0.457	0.448	0.576	0.275
Kardex utilization	0.079	0.079	0.079	0.078	0.078	0.079
Orders completed (5 days)	1943	1942	1942	1942	1931	1943